

10.8 GREEN'S FUNCTIONS, THE $i\epsilon$ AND THE FEYNMAN PROPAGATOR

LecNote6 evaluates the oscillator path integral exactly,

$$\begin{aligned}\langle 0|0\rangle_f &= \exp\left[\frac{i}{2} \int dt dt' f(t) G(t-t') f(t')\right], \\ G(t) &= \int \frac{dE}{2\pi} \frac{e^{-iEt}}{-E^2 + \omega^2 - i\epsilon} = \frac{i}{2\omega} e^{-i\omega|t|},\end{aligned}\tag{10.8.1}$$

recognizes G as a Green's function, $(\partial_t^2 + \omega^2)G(t) = \delta(t)$ (p. 3), and reads off $\langle 0|TQ(t_1)Q(t_2)|0\rangle = G(t_1 - t_2)/i$ (p. 4). That leaves three questions. LecNote4 calls the states made by $a^\dagger$ particles; what are the particles in a quantum-mechanics problem that has only one? Why is G a Green's function, and why this one, which is neither the retarded nor the advanced one? And how are the $(1 - i\epsilon)H$ of LecNote5 p. 8, the pole placement on LecNote6 p. 3 and the independence of the result from the initial and final states related? In short: the particles are the quanta of the oscillator. Being a Green's function follows from the equation of motion, in any state; being *this* Green's function is a statement about the vacuum at both ends. The projection onto the vacuum, the pole placement and the Wick rotation are one prescription. The note opens with Green's functions in general, and is the one-mode version of the source discussion in §10.6, Eqs. (10.6.14)–(10.6.16).

GREEN'S FUNCTIONS: FROM IMPULSE RESPONSE TO PROPAGATORS

A Green's function solves an inhomogeneous linear differential equation with a point-like source. It is the response of a system to an idealized impulse, and its interpretation changes from classical mechanics to quantum field theory, tied each time to causality.

Classical impulse response. For a driven oscillator, $\ddot{x}(t) + \omega_0^2 x(t) = F(t)/m$, the Green's function $G(t, t')$ is the position at time t produced by a unit impulse at t' . The **retarded** Green's function enforces causality, so that the effect does not precede the cause:

$$G_R(t, t') = \begin{cases} \frac{1}{m\omega_0} \sin(\omega_0(t - t')) & t \geq t', \\ 0 & t < t'. \end{cases}$$

The solution for an arbitrary force is the convolution $x(t) = \int dt' G_R(t, t') F(t')$, the sum of the responses to a continuous series of impulses.

1272 *The quantum-mechanical kernel.* In quantum mechanics the Green's
 1273 function becomes the **propagator**, or Schrödinger kernel,
 1274 $K(x, t; x', t') = \langle x | e^{-iH(t-t')} | x' \rangle$ (with $\hbar = 1$). It is the amplitude for a particle
 1275 to go from (x', t') to (x, t) , and, multiplied by $\theta(t - t')$, it solves the
 1276 Schrödinger equation with a delta-function source:
 1277 $(i\partial_t - H_x) [\theta(t - t')K] = i \delta(t - t') \delta(x - x')$. It evolves any initial wavefunction,

$$\Psi(x, t) = \int dx' K(x, t; x', t') \Psi(x', t'),$$

1278 and it is the starting point of the path-integral formulation of quantum
 1279 mechanics. This kernel is not the same object as the propagator of
 1280 quantum field theory. The kernel is a Green's function of $i\partial_t - H$ and
 1281 evolves the wavefunction. The field-theory propagator is a Green's
 1282 function of the Klein–Gordon operator, built from two-point correlation
 1283 functions of the field, and it is the subject of the rest of this note.
 1284 *Causality in quantum field theory.* In relativistic QFT, which must
 1285 account for antiparticles, two Green's functions of the same operator play
 1286 different roles.

- 1287 • **The retarded function** G_R , as in the classical case, vanishes for
 1288 $t < t'$ and describes the response of the field to a source: with a
 1289 source J , $\langle \phi \rangle = \int G_R J$. It is built from the commutator of the field
 1290 at the two points, which for a free field is a c-number, the same in
 1291 every state. It embodies the classical notion of cause and effect; it
 1292 is a response, not a propagation amplitude.
- 1293 • **The Feynman propagator** G_F describes two processes in one
 1294 function: a particle propagating forward in time ($t > t'$), and an
 1295 antiparticle propagating forward in time, which is mathematically
 1296 the same as a particle propagating *backward* in time ($t < t'$).

1297 The Feynman prescription is a rule for passing the poles in the complex
 1298 energy plane that yields this. It is what Feynman diagrams use, where an
 1299 internal line can carry a particle or an antiparticle. The Feynman
 1300 propagator gives up the purely retarded boundary condition (it is a
 1301 time-ordered product, not a retarded commutator) in order to describe a
 1302 quantum theory with particle creation and annihilation. The rest of this
 1303 note shows, for a single mode, why the vacuum selects it and why the
 1304 retarded function is a response rather than an amplitude.

1305 THE OSCILLATOR

1306 Conventions are those of LecNote6: $\hbar = 1$, unit mass, $H = \frac{1}{2}P^2 + \frac{1}{2}\omega^2 Q^2$ with
 1307 the ground-state energy shifted to $E_0 = 0$ (LecNote5 p. 8), the source
 1308 enters the Lagrangian as $+fQ$ (LecNote5 p. 7) with f real, and

1309 $\tilde{f}(E) = \int dt e^{iEt} f(t)$. Without the source, in the Heisenberg picture,

$$Q(t) = \frac{1}{\sqrt{2\omega}} (a e^{-i\omega t} + a^\dagger e^{i\omega t}), \quad [a, a^\dagger] = 1, \quad H = \omega a^\dagger a. \quad (10.8.2)$$

1310 THE QUANTA ARE THE PARTICLES

1311 In LecNote5 there is one particle, and Q is its position. Its number is one,
 1312 trivially and forever. That is not the particle of LecNote4. In LecNote7 the
 1313 coordinate $Q(t)$ becomes the field $\varphi(\mathbf{x}, t)$, and each mode of the field is an
 1314 oscillator like this one, with $\omega \rightarrow \omega_{\mathbf{k}}$ (LecNote7 p. 1). The particles of
 1315 LecNote4, created by $a^\dagger(\mathbf{k})$, are the quanta of those oscillators. So
 1316 Srednicki's §7 is best read as a field theory in $0 + 1$ dimensions: Q is the
 1317 field, $a^\dagger$ creates one quantum of energy ω , and $|n\rangle$ is the n -particle state.
 1318 Read this way, the oscillator is a *free* theory. The potential $\frac{1}{2}\omega^2 Q^2$ is the
 1319 mass term, not an interaction: the action is quadratic. $N = a^\dagger a$ commutes
 1320 with H , so the number of particles is conserved, and the only time
 1321 dependence is the phase $e^{-i\omega t}$ that each quantum carries. That is free
 1322 propagation, which is all a free particle does. The analogue of the
 1323 normalization $\langle p | \varphi(0) | 0 \rangle = 1$ (LecNote4 p. 8) is

$$\langle 1 | Q(0) | 0 \rangle = \frac{1}{\sqrt{2\omega}}, \quad (10.8.3)$$

1324 which is exactly 1 when the one-particle state is normalized as on
 1325 LecNote4 p. 1, $|k\rangle = \sqrt{2\omega} |1\rangle$.

1326 What does change the number of particles is the source. The term
 1327 $fQ = f(a + a^\dagger)/\sqrt{2\omega}$ emits and absorbs quanta: it is the external
 1328 interaction, and $\langle 0 | 0 \rangle_f$ is the amplitude that the source, switched on and
 1329 off, leaves no quantum behind. A self-interaction such as Q^3 or Q^4
 1330 contains $a^{\dagger 3}$, $a^{\dagger 4}$, $\dots$, and with it N is no longer conserved.

1331 Eq. (10.8.1) gives the probability directly. Since $|e^{iX/2}|^2 = e^{-\text{Im } X}$ and
 1332 $\text{Im } G(t) = \cos(\omega t)/2\omega$,

$$|\langle 0 | 0 \rangle_f|^2 = \exp \left[-\frac{|\tilde{f}(\omega)|^2}{2\omega} \right]. \quad (10.8.4)$$

1333 The source leaves the oscillator in a coherent state, with a
 1334 Poisson-distributed number of quanta of mean $\bar{n} = |\tilde{f}(\omega)|^2/2\omega$, and
 1335 (10.8.4) is $P(0) = e^{-\bar{n}}$. Only the component of f at frequency ω enters: a
 1336 quantum is emitted only if the source can supply its energy. Eq. (10.6.15)
 1337 is the same statement for a field. As a check, take $f = 0.8 e^{-t^2/2s^2}$ with
 1338 $s = 0.8$ and $\omega = 1$, so $\bar{n} = 0.6785$. Evolving $|0\rangle$ with $H - fQ$ in a truncated
 1339 Fock space gives $\langle N \rangle = 0.6785$ afterwards, occupation probabilities 0.5074,
 1340 0.3443, 0.1168, 0.0264 for $n = 0, \dots, 3$ (the Poisson values), and
 1341 $\langle 0 | U | 0 \rangle = 0.65981 + 0.26837 i$, which is (10.8.1) to all the digits shown.

1342 EVERY TIME-ORDERED TWO-POINT FUNCTION IS A 1343 GREEN'S FUNCTION

The Green's-function property on LecNote6 p. 3 needs neither the path integral nor the vacuum. Take any normalized state $|\psi\rangle$. Differentiating the θ -functions of the time ordering gives delta functions times equal-time commutators. With $\dot{Q} = P$, $\dot{P} = -\omega^2 Q$ and $[Q(t), P(t)] = i$,

$$\begin{aligned}\partial_t \langle \psi | T Q(t) Q(t') | \psi \rangle &= \langle \psi | T P(t) Q(t') | \psi \rangle + \delta(t - t') \langle \psi | [Q(t), Q(t')] | \psi \rangle, \\ (\partial_t^2 + \omega^2) \langle \psi | T Q(t) Q(t') | \psi \rangle &= \delta(t - t') \langle \psi | [P(t), Q(t)] | \psi \rangle = -i \delta(t - t').\end{aligned}\quad (10.8.5)$$

1344 So $i \langle \psi | T Q(t) Q(t') | \psi \rangle$ is a Green's function of the oscillator, whatever $|\psi\rangle$ is.
1345 With a potential $V(Q)$ added, the right-hand side gains
1346 $-\langle \psi | T V'(Q(t)) Q(t') | \psi \rangle$, but the contact term $-i \delta(t - t')$ is always there. This
1347 is the simplest Schwinger–Dyson equation; Eq. (10.6.21) is its
1348 field-theory version.

For a number eigenstate, (10.8.2) gives

$$\begin{aligned}\langle n | T Q(t) Q(0) | n \rangle &= \frac{G(t)}{i} + \frac{n \cos \omega t}{\omega}, \\ \int dt e^{iEt} \langle n | T Q(t) Q(0) | n \rangle &= \frac{i}{E^2 - \omega^2 + i\epsilon} + 2\pi n \delta(E^2 - \omega^2).\end{aligned}\quad (10.8.6)$$

1349 The extra term solves the homogeneous equation, so (10.8.5) still holds,
1350 but this is a different Green's function. In energy it is an on-shell delta
1351 function. In a thermal state n is replaced by the Bose factor, and (10.8.6)
1352 becomes the time-ordered component of the real-time propagator at finite
1353 temperature.

The same goes for the commutator. For this H , $[Q(t), Q(0)] = -i \sin(\omega t)/\omega$ is a c-number, the same in every state, and both

$$\begin{aligned}G_R(t) &= i\theta(t) \langle \psi | [Q(t), Q(0)] | \psi \rangle = \theta(t) \frac{\sin \omega t}{\omega}, \\ G_A(t) &= -i\theta(-t) \langle \psi | [Q(t), Q(0)] | \psi \rangle = -\theta(-t) \frac{\sin \omega t}{\omega}\end{aligned}\quad (10.8.7)$$

1354 satisfy $(\partial_t^2 + \omega^2)G = \delta$.

1355 The equation of motion fixes a Green's function $G(t - t')$ only up to
1356 $Ae^{i\omega t} + Be^{-i\omega t}$. Being a Green's function is the dynamics. Which Green's
1357 function is a boundary condition. G_R and G_A are the same in every state;
1358 for the time-ordered function the boundary condition is the state.

1359 WHICH ONE: THE FEYNMAN GREEN'S FUNCTION

1360 In energy the three Green's functions differ only in which way the poles
1361 at $E = \pm\omega$ are pushed off the real axis (table on p. 53, Fig. 10.2). The

1362 retarded function closes below for $t > 0$ and picks up both poles; for $t < 0$
 1363 it closes above and finds none. The Feynman function picks up $+\omega$ for
 1364 $t > 0$ and $-\omega$ for $t < 0$.

	poles	$G(t)$	operator form
retarded	$\pm\omega - i\epsilon$	$\theta(t) \sin(\omega t)/\omega$	$i\theta(t)\langle\psi [Q(t), Q(0)] \psi\rangle$, any ψ
advanced	$\pm\omega + i\epsilon$	$-\theta(-t) \sin(\omega t)/\omega$	$-i\theta(-t)\langle\psi [Q(t), Q(0)] \psi\rangle$, any ψ
Feynman	$\omega - i\epsilon, -\omega + i\epsilon$	$(i/2\omega) e^{-i\omega t }$	$i\langle 0 TQ(t)Q(0) 0\rangle$, vacuum only

THE THREE GREEN'S FUNCTIONS OF $\partial_t^2 + \omega^2$

The three Green's functions met in these notes. All satisfy $(\partial_t^2 + \omega^2)G = \delta(t)$.

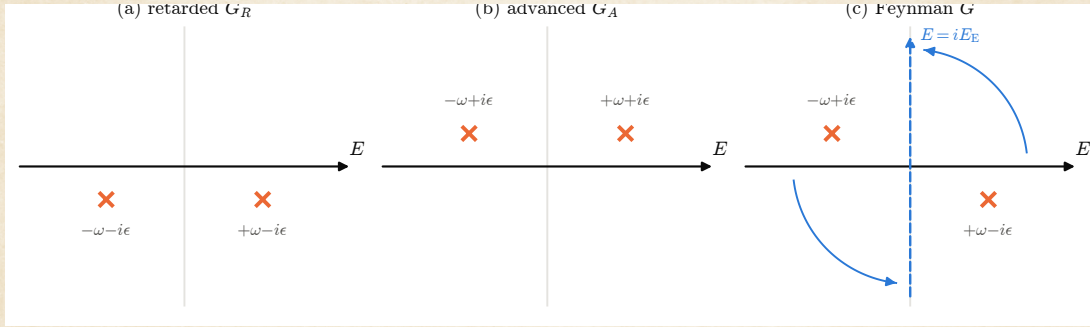


Figure 10.2: Pole placements in the complex E plane. (c) also shows the Wick rotation $E = iE_E$, which sweeps the first and third quadrants. It crosses no pole only for the Feynman placement.

1365 *Reading it off the operators.* From (10.8.2), $\langle 0|Q(t)Q(t')|0\rangle = e^{-i\omega(t-t')}/2\omega$.
 1366 The right-hand Q creates a quantum at t' , the left-hand Q absorbs it at t ,
 1367 and in between the state carries energy ω . Time ordering puts the later
 1368 time on the left, so whichever of t, t' is earlier, the quantum is created
 1369 then:

$$G(t) = \frac{i}{2\omega} [\theta(t) e^{-i\omega t} + \theta(-t) e^{+i\omega t}]. \quad (10.8.8)$$

1370 Positive frequency propagates forward in time and negative frequency
 1371 backward. Equivalently, positive energy always flows from the earlier
 1372 time to the later one. The vacuum is the bottom of the spectrum, so an
 1373 operator acting on it can only add energy; it can never borrow any. That
 1374 is the content of the Feynman prescription. For a complex field the
 1375 backward-going negative-frequency part is read as an antiparticle going
 1376 forward, as in the overview above. For the real Q here it is the same
 1377 quantum, created at the earlier of the two times.

Three decompositions. From (10.8.1) and (10.8.7),

$$G(t) = G_R(t) + \frac{i}{2\omega} e^{i\omega t} = G_A(t) + \frac{i}{2\omega} e^{-i\omega t}, \quad (10.8.9)$$

$$\text{Re } G(t) = \frac{1}{2} [G_R(t) + G_A(t)] = \frac{\sin \omega|t|}{2\omega}, \quad \text{Im } G(t) = \frac{1}{2} \langle 0|\{Q(t), Q(0)\}|0\rangle = \frac{\cos \omega t}{2\omega}. \quad (10.8.10)$$

1378 The Feynman function is the retarded one plus a solution of the free
 1379 equation. Its real part is built from the commutator alone, so it is the
 1380 same in every state: it is the dynamics. Its imaginary part is the
 1381 symmetrized vacuum correlation; at $t = 0$ it is $\langle 0|Q^2|0\rangle = 1/2\omega$, the
 1382 zero-point spread. In $|n\rangle$ it becomes $(2n + 1) \cos(\omega t)/2\omega$, which is (10.8.6)
 1383 again. So the Feynman function is the dynamics plus the vacuum's
 1384 fluctuations, and the state sits entirely in the homogeneous part. In
 1385 energy, $1/(-E^2 + \omega^2 - i\epsilon) = \mathcal{P} 1/(\omega^2 - E^2) + i\pi \delta(E^2 - \omega^2)$: the imaginary part
 1386 is on shell, and it is what makes $|\langle 0|0\rangle_f| < 1$ in (10.8.4).

1387 AN AMPLITUDE, NOT A RESPONSE

1388 Why is the answer not the retarded function, the one classical mechanics
 1389 uses? Because $\langle 0|0\rangle_f$ answers a different question.
 1390 *Response.* Start in $|0\rangle$ and push the oscillator with f . With $H - fQ$ the
 1391 Heisenberg equation $\ddot{Q} + \omega^2 Q = f$ holds as an operator statement, so the
 1392 expectation value obeys the classical equation, with $\langle Q\rangle = \langle P\rangle = 0$ in the
 1393 past:

$$\langle Q(t)\rangle = \int dt' G_R(t - t') f(t'). \quad (10.8.11)$$

1394 It is real and causal. It is an in-in quantity: the same state on both sides.
 1395 *Amplitude.* $\langle 0|0\rangle_f = \langle 0_{\text{out}}|0_{\text{in}}\rangle_f$ is the amplitude to start in the vacuum *and*
 1396 *end* in the vacuum (§10.6). Its first derivative is an in-out matrix
 1397 element,

$$q_{\text{io}}(t) \equiv \frac{1}{i} \frac{\delta \ln \langle 0|0\rangle_f}{\delta f(t)} = \frac{\langle 0_{\text{out}}|Q(t)|0_{\text{in}}\rangle_f}{\langle 0_{\text{out}}|0_{\text{in}}\rangle_f} = \int dt' G(t - t') f(t'), \quad (10.8.12)$$

1398 the one-mode version of Eq. (10.6.16). It is complex, and it is not zero
 1399 before the source is switched on. That is not acausal: q_{io} is conditioned
 1400 on ending in the vacuum, and it is not an observable. The vacuum is
 1401 imposed at both ends, so this is a boundary-value problem with
 1402 conditions at $t \rightarrow -\infty$ *and* $t \rightarrow +\infty$, not an initial-value problem. By
 1403 (10.8.8), q_{io} contains only $e^{-i\omega t}$ after the source has acted and only $e^{+i\omega t}$
 1404 before. Those are the two conditions. With the source on, define
 1405 $a(t) = e^{i\omega t} [\sqrt{\omega/2} Q(t) + iP(t)/\sqrt{2\omega}]$, which is a_{in} before the source and a_{out}
 1406 after it. Then $a_{\text{in}}|0_{\text{in}}\rangle = 0$ removes the a part early, and $\langle 0_{\text{out}}|a_{\text{out}}^\dagger = 0$
 1407 removes the $a^\dagger$ part late.

1408 Fig. 10.3 and the table on p. 55 show both for a Gaussian pulse. The
 1409 operator columns evolve $|0\rangle$ forward and $\langle 0|$ backward in a truncated Fock
 1410 space, which does by hand what the $i\epsilon$ does in the path integral. After the
 1411 pulse, $\langle Q\rangle = 2 \text{Re } q_{\text{io}}$ (to 5×10^{-7}): the in-out quantity keeps only the
 1412 positive-frequency half of the response. Before the pulse $\langle Q\rangle = 0$, but
 1413 $q_{\text{io}} = (i/2\omega) e^{i\omega t} \tilde{f}(\omega)^* \neq 0$.

ωt	in-in $\langle Q(t) \rangle$		in-out $q_{\text{io}}(t)$	
	operators	$\int G_R f$	operators	$\int G f$
-3.0	0.00000	0.00000	$0.00514 - 0.03604 i$	$0.00514 - 0.03604 i$
0.0	0.02597	0.02597	$0.02597 + 0.03640 i$	$0.02597 + 0.03640 i$
1.5	0.07351	0.07351	$0.03719 + 0.00258 i$	$0.03719 + 0.00258 i$
4.0	-0.05510	-0.05510	$-0.02755 - 0.02380 i$	$-0.02755 - 0.02380 i$
7.0	0.04783	0.04783	$0.02392 + 0.02744 i$	$0.02392 + 0.02744 i$

RESPONSE AND IN-OUT VALUE FOR A GAUSSIAN PULSE

$f = 0.05 e^{-t^2/2s^2}$, $s = 0.8$, $\omega = 1$. “Operators”: time evolution with $H - fQ$ in an 80-state Fock space. The convolutions use (10.8.7) and (10.8.1).

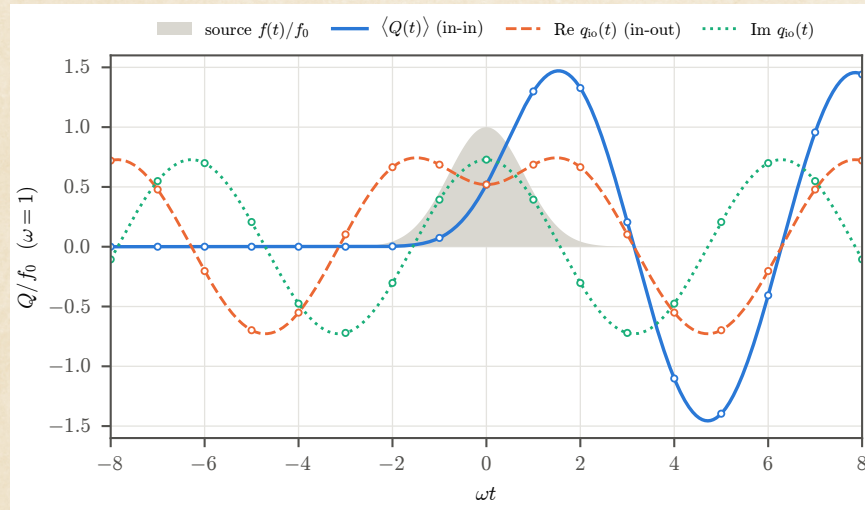


Figure 10.3: The pulse of the table on p. 55. Lines: operator calculation. Circles: $\int G_R f$ and $\int G f$. The response $\langle Q \rangle$ is zero until the pulse arrives; the in-out value is not.

1414 So G should not be read as the way the position responds to a kick. At
 1415 $f = 0$ both G and G_R are vacuum correlators, a time-ordered product and
 1416 a retarded commutator. With a source on, G is the kernel of the in-out
 1417 amplitude and G_R is the kernel of the response. LSZ needs the in-out
 1418 kernel: LecNote4 builds the S -matrix from vacuum-to-vacuum
 1419 time-ordered products. Expectation values in a driven state call for an
 1420 in-in (closed-time-path, or Schwinger–Keldysh) generating functional,
 1421 which this course does not use.

1422 THE $(1 - i\epsilon)$ PRESCRIPTION: ONE $i\epsilon$, THREE FACES

1423 The factor $1 - i\epsilon$ is put into the Hamiltonian (LecNote5 p. 8) for two reasons:
 1424 it makes the oscillatory path integral converge, by damping high-energy
 1425 configurations with $e^{-\epsilon HT}$, and it projects the initial and final states onto
 1426 the vacuum, so that the path integral computes the vacuum-to-vacuum
 1427 amplitude whatever states were actually prepared. Replacing $H \rightarrow (1 - i\epsilon)H$
 1428 in e^{-iHT} is the same as $T \rightarrow T(1 - i\epsilon)$: time is tilted slightly into the lower
 1429 half plane. That one change does three things, which are the same thing.
 1430 *It projects onto the vacuum.* With $c_n = \langle n | \psi \rangle$ and elapsed time $T \rightarrow +\infty$, on
 1431 the tilted line

$$e^{-iHT(1-i\epsilon)}|\psi\rangle = \sum_n c_n e^{-iE_n T} e^{-\epsilon E_n T} |n\rangle \rightarrow c_0 |0\rangle \quad (10.8.13)$$

1432 as $T \rightarrow \infty$ at fixed ϵ , i.e. for $\epsilon T \gg 1$, with $\epsilon \rightarrow 0$ taken last (LecNote5 p. 8).
 1433 The same happens at the far end. The two overlaps with $|0\rangle$, which must
 1434 both be nonzero, are removed by $\langle 0 | 0 \rangle_{f=0} = 1$. This needs a unique ground
 1435 state; with a degenerate one the projection is onto the ground subspace,
 1436 and the result does depend on the preparation. A gap makes the
 1437 approach exponential, $e^{-\epsilon \omega T}$ for the oscillator.

1438 *It places the poles.* In the path integral the same tilt multiplies $\frac{1}{2}\dot{q}^2$ by $1 + i\epsilon$
 1439 and $\frac{1}{2}\omega^2 q^2$ by $1 - i\epsilon$ (LecNote6 p. 1), so the kernel is
 1440 $(1 + i\epsilon)E^2 - (1 - i\epsilon)\omega^2 = E^2 - \omega^2 + i\epsilon(E^2 + \omega^2)$. The coefficient of ϵ is positive,
 1441 so this is $E^2 - \omega^2 + i\epsilon$ (p. 2), with poles at $E = \pm(\omega - i\epsilon')$, $\epsilon' = \epsilon/2\omega$ after the
 1442 rescaling: the Feynman placement. The same can be seen in time. On
 1443 the tilted line $|G(t(1 - i\epsilon))| = e^{-\epsilon \omega |t|}/2\omega$ decays in both directions, while G_R
 1444 grows like $e^{\epsilon \omega t}$ for $t > 0$ and G_A like $e^{\epsilon \omega |t|}$ for $t < 0$. “The vacuum at both
 1445 ends” and “the Feynman poles” are the same requirement: the Green’s
 1446 function that stays bounded when time is tilted.

1447 *It is the start of the Wick rotation.* Tilt all the way, $t = -i\tau$. Then
 1448 $\partial_t^2 + \omega^2 \rightarrow -\partial_\tau^2 + \omega^2$, a positive operator, and its Green’s function has no
 1449 freedom left:

$$(-\partial_\tau^2 + \omega^2) G_E(\tau) = \delta(\tau), \quad G_E(\tau) = \frac{e^{-\omega|\tau|}}{2\omega}, \quad G(-i\tau) = i G_E(\tau). \quad (10.8.14)$$

Adding $Ae^{\omega\tau} + Be^{-\omega\tau}$ would make G_E blow up at one end, so the bounded Euclidean Green's function is unique, and G is its analytic continuation. In energy, the contour can be rotated to $E = iE_E$ without crossing a pole only when the poles sit in the second and fourth quadrants (Fig. 10.2c; the pole sketch on LecNote6 p. 3 has them there). With the retarded or advanced placement one pole is always in the way.

It is tempting to read the logic backwards: the result is a Green's function, so it cannot depend on how the states were prepared, and that justifies the $i\epsilon$. But every state gives a Green's function, (10.8.5). Being a Green's function is not what removes the preparation; the projection does. The projection is the $i\epsilon$, and the same $i\epsilon$, seen in energy, is the contour. Nor is independence of the preparation independence of the state: the $i\epsilon$ replaces whatever was prepared by the vacuum, for any preparation that overlaps it. With $|n\rangle$ at both ends the answer would be the different Green's function (10.8.6).

Adiabatic switching. In scattering theory, interactions are often said to be switched off adiabatically at early and late times to define in and out states. The $i\epsilon$ is related to this but is not the same thing. It does not switch the interaction off: it multiplies all of H , free and interacting parts alike, and projects onto the ground state of the full Hamiltonian, the interacting vacuum. Adiabatic switching reaches the same state from the other side: by the Gell-Mann–Low theorem, switching the interaction on slowly carries the free vacuum into an eigenstate of the full Hamiltonian, when that limit exists.

What it does for scattering.

- **Vacuum selection.** Only the ground state survives at both ends, so the result does not depend on the states prepared, provided they overlap with the vacuum.
- **Vacuum-to-vacuum correlators.** With the vacuum at both ends, the path integral with sources computes $Z[J] = \langle 0|0\rangle_J$ and, by differentiation, vacuum expectation values of time-ordered products, with the Feynman propagator as the free two-point function.
- **LSZ.** The LSZ formula expresses S -matrix elements through exactly these time-ordered vacuum correlators (§10.6). The particles of the in and out states enter through LSZ; the $i\epsilon$ supplies the vacuum they are built on.

FROM ONE OSCILLATOR TO THE FREE FIELD

In LecNote7 each mode of φ is an oscillator with $\omega_k = \sqrt{k^2 + m^2}$, and the same steps give the Feynman propagator

$\Delta(x - x') = \int \frac{d^4k}{(2\pi)^4} e^{ik(x-x')} / (k^2 + m^2 - i\epsilon)$, whose k^0 integral is (10.8.1) with

$\omega \rightarrow \omega_{\mathbf{k}}$ (LecNote7 p. 3). Everything above holds mode by mode. The quanta of mode $\mathbf{k}$ are particles of momentum $\mathbf{k}$. $|\langle 0|0\rangle_J|^2$ is the probability that the source emits nothing, and $-\ln |\langle 0|0\rangle_J|^2$ is the mean number it emits (Eq. (10.6.15)). $\text{Re } \Delta$ comes from the commutator and $\text{Im } \Delta$ from the vacuum's fluctuations. The in-in response to J is the retarded function, and the in-out value is $\int \Delta J$.
 LecNote7 p. 1 writes the prescription as $m^2 \rightarrow (1 - i\epsilon)m^2$. In fact $(1 - i\epsilon)H_0$ damps all three terms of H_0 and gives the kernel $-(k^2 + m^2) + i\epsilon(k_0^2 + \mathbf{k}^2 + m^2)$, whose ϵ coefficient is positive. So the general form is $\omega_{\mathbf{k}}^2 \rightarrow \omega_{\mathbf{k}}^2 - i\epsilon$ in every mode, i.e. $m^2 \rightarrow m^2 - i\epsilon$. Writing $(1 - i\epsilon)m^2$ is equivalent for $m \neq 0$ but gives nothing at $m = 0$, where $k^2 - i\epsilon$ is still the right prescription.
 At $m = 0$ there is no gap. In quantum mechanics the limit $\omega \rightarrow 0$ is a free particle, which has no normalizable ground state, and $G \propto 1/\omega$ diverges. In the massless field $\omega_{\mathbf{k}} \rightarrow 0$ only as $\mathbf{k} \rightarrow 0$, and in $3 + 1$ dimensions the soft modes carry little weight: $\int d^3k/2\omega_{\mathbf{k}} \sim \int k dk$ converges, so Δ has no infrared divergence. (In $1 + 1$ dimensions it does.) The projection onto the Fock vacuum still works, but its approach is a power of T rather than exponential.

SUMMARY

- The particles of the oscillator are its quanta. The oscillator is the free theory of one mode, and the source f is the external interaction that emits and absorbs them.
- Every time-ordered two-point function is a Green's function, in any state. That is the equation of motion. Which Green's function it is depends on the state.
- The vacuum at both ends picks the Feynman function: positive energy flows forward in time, $G = G_R + (i/2\omega)e^{i\omega t}$, $\text{Re } G$ is the dynamics and $\text{Im } G$ the vacuum's fluctuations.
- With a source on, G is the kernel of the in-out amplitude, and G_R is the kernel of the response of $\langle Q \rangle$. LSZ needs the former.
- $(1 - i\epsilon)H$ tilts time. The projection onto the vacuum, the Feynman poles and the Wick rotation are one prescription.